

The world's hunger for processing power is insatiable, and for storage capacity and speed, even more so. Storage devices—both volatile, as in RAM, and non-volatile, as in hard disks—are always getting larger and faster. New technologies and methodologies seem to be forever on the horizon. Recall that when Intel released the original Pentium, it was believed by many that a limit on the number of transistors that could be crammed onto a chip was being approached. Will there ever be some sort of fundamental impediment to further development?

Not if research continues at the current rate, driven by worldwide, exponentially ever-increasing demand. Stor-

happened largely due to increases in areal density, that is, how many bits are crammed into, say, a square inch. A lot of research in storage focuses on how to improve areal, and therefore data, densities. However, in a couple of years, a limit on the areal density that disks can achieve will be reached. This limit will be due to the superparamagnetic effect.

Bits on a hard disk are encoded by changing a magnetic property of the disk material—the spin. The superparamagnetic effect (SPE) happens when bits are crammed too close together. When this happens, the energy holding the magnetic spin in the atoms making up a bit becomes smaller, and comparable to the thermal energy of the atoms. This means

perpendicular. Of these, the last—called perpendicular recording—holds the most promise, and is widely believed to be the recording technology of the future. The concept is simple: in conventional-longitudinal-recording, the magnetisation of the bits is parallel to the direction of the movement of the head. With perpendicular recording, the bit magnetisations would be perpendicular to the head movement.

Dr Mark Kryder, Director of Seagate Research, says there's time before longitudinal recording will need to be widely employed. According to Dr Kryder, the switch to perpendicular recording will occur at an areal density of somewhere between 100 and 200 gigabits per square

WHAT'S in Store?

...for hard drives and memory

age research is looking at everything—from materials to fabrication methods, to novel techniques. And every once in a while, something comes up that takes performance up by an order of magnitude.

It seems that the industry will continue to deliver—you'll have a terabyte hard disk a few years from now, and a pentabyte one a decade later, with RAM to match.

Hard disks

The hard disk, the mainstay of personal and corporate storage, has faithfully obeyed the exponential law. This has

that the bits may get demagnetised—they may randomly change (flip) from zeroes to ones, or vice-versa, leading to data corruption. The SPE may become important as early as next year.

Perpendicular recording

Several techniques are being looked into to overcome the SPE, and therefore allow one to cram more bits together, while preventing them from flipping. These include using different magnetic materials more resistant to the SPE, using an entirely different kind of material such as plastic, and changing the bit orientation on the disk from longitudinal to

inch. This switch would open up the potential for unprecedented areal density increases. He says, "We estimate that perpendicular recording methods may take us all the way to one terabit per square inch. When that level is reached, a single 3.5-inch disk will store over one terabyte of information."

OAW technology

Quinta Corporation, a subsidiary of Seagate, combated the SPE limit in 1999 with its Optically Assisted Winchester (OAW) technology. In OAW, instead of using a regular head to read and write data off a hard disk platter, a laser is

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